

Technical Requirements Document (TRD)

NexGenius Solutions Website

Version: 1.0

Document Type: Technical Requirements Document (TRD)

Project: NexGenius Solutions Website

Based On: Product Requirements Document (PRD)

1. Project Overview

Objective

Develop a fast, scalable, SEO-friendly, and responsive company website for NexGenius Solutions that showcases AI services, captures leads, and supports both Pakistani and international clients.

2. System Architecture

Client (Browser)

↓

Next.js Frontend

↓

Next.js API Routes

↓

Email Service (Resend / Brevo)

↓

Company Email Inbox

External Services

- WhatsApp

- Google Analytics
 - Google Search Console
 - Microsoft Clarity
-

3. Technology Stack

Frontend

- Next.js 15 (App Router)
 - React 19
 - TypeScript
 - Tailwind CSS
 - shadcn/ui
 - Framer Motion
 - Lucide Icons
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Backend

For MVP:

- Next.js API Routes
- Server Actions (where appropriate)

No separate Express/NestJS backend required.

Future:

- NestJS or FastAPI (if CRM, authentication, or dashboards are added)
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4. Folder Structure

```
app/  
components/  
features/  
lib/  
hooks/  
types/  
public/  
styles/  
content/
```

```
app/api/contact/  
app/api/blog/
```

5. Pages

Home

Route

/

Sections

- Hero
 - Trusted Technologies
 - Services
 - Why Choose Us
 - Process
 - CTA
 - Footer
-

About

/about

Sections

- Company
 - Vision
 - Mission
 - Values
 - Team (Future)
-

Services

/services

Each service should have:

- Hero

- Description
 - Benefits
 - Technologies
 - CTA
-

Blog

/blog

Features

- Blog List
- Categories
- Search
- Pagination
- Related Articles

Dynamic Route

/blog/[slug]

Contact

/contact

Includes

- Contact Form
 - Contact Information
 - WhatsApp CTA
-

6. Contact Form

Fields

- Name
- Company
- Email
- Phone
- Country
- Service
- Budget

- Message

Validation

- Required Fields
- Email Validation
- Phone Validation
- Spam Protection

Submission Flow

User



Frontend Validation



API Route



Email Service



Company Email



Success Response

7. Email Integration

Provider

Recommended:

- Resend

Alternative

- Brevo

Email Content

- Name
- Company
- Email
- Phone
- Country
- Selected Service
- Budget
- Message
- Timestamp

Admin receives notification instantly.

8. WhatsApp Integration

Floating Button

Functions

- Open WhatsApp
- Pre-filled Message

Example

Hello NexGenius Solutions,

I'm interested in your AI services.

I'd like to schedule a consultation.

9. Blog System

Initial Version

Static MDX Blog

Future Version

CMS

Recommended CMS

- Sanity
- Contentful
- Strapi

Features

- SEO
- Categories
- Tags
- Reading Time
- Featured Image
- Author

10. SEO Requirements

Every page must include

- Unique Title
- Meta Description
- Canonical URL
- Open Graph Tags
- Twitter Cards
- Structured Data (Schema.org)

Files

- sitemap.xml
- robots.txt

11. Performance Requirements

Target

Lighthouse

Performance ≥ 95

Accessibility ≥ 95

Best Practices ≥ 100

SEO ≥ 100

Page Load

Less than 2 seconds

Largest Contentful Paint

Under 2.5 seconds

12. Responsive Design

Support

Desktop

Laptop

Tablet

Mobile

No horizontal scrolling.

13. UI Components

Reusable Components

- Navbar
 - Footer
 - Buttons
 - Cards
 - Service Cards
 - Blog Cards
 - Contact Form
 - CTA Banner
 - FAQ
 - Testimonials (Future)
-

14. Animations

Library

Framer Motion

Animations

- Fade
- Slide
- Scale
- Hover
- Scroll Reveal

Animations should remain lightweight.

15. Security

Environment Variables

Store

- Email API Key
- Analytics Keys

Protection

- reCAPTCHA
 - Rate Limiting
 - Input Validation
 - HTTPS
-

16. Accessibility

WCAG 2.1

Requirements

- Proper Heading Structure
- Keyboard Navigation
- Alt Text
- Focus States

- Color Contrast
-

17. Analytics

Google Analytics 4

Track

- Visitors
- Contact Form Submissions
- CTA Clicks
- WhatsApp Clicks
- Blog Views

Microsoft Clarity

Track

- Heatmaps
 - Session Recording
-

18. Deployment

Hosting

Vercel

Domain

nexgeniussolutions.com

Environment Variables

Configured inside Vercel.

19. Future Enhancements

Phase 2

- Authentication

- Admin Dashboard
 - CRM Integration
 - Blog CMS
 - Client Portal
 - AI Quote Generator
 - AI Website Audit Tool
 - Newsletter
 - Live Chat
 - Multi-language Support
-

20. Non-Functional Requirements

Availability

99.9%

Scalability

Support future API integrations without major refactoring.

Maintainability

Component-based architecture.

Code Quality

- ESLint
 - Prettier
 - TypeScript Strict Mode
-

21. Acceptance Criteria

- Responsive across all devices
 - Contact form successfully sends emails
 - WhatsApp integration works correctly
 - All pages pass SEO validation
 - Lighthouse score ≥ 95
 - No console errors
 - Optimized images and assets
 - Clean, modular, and documented codebase
 - Ready for deployment on Vercel
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Final Technical Summary

The NexGenius Solutions website will be built using a modern Next.js architecture with a lightweight backend powered by API Routes. The application will prioritize speed, SEO, scalability, and maintainability while integrating email notifications, WhatsApp communication, analytics, and a blog system. The technical foundation will be designed to support future expansion into a full AI platform with dashboards, CRM integrations, and advanced business automation features.